Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 3

Data visualization using line, bar and pie charts

Experiment 3.a: Show data visualization using a line plot

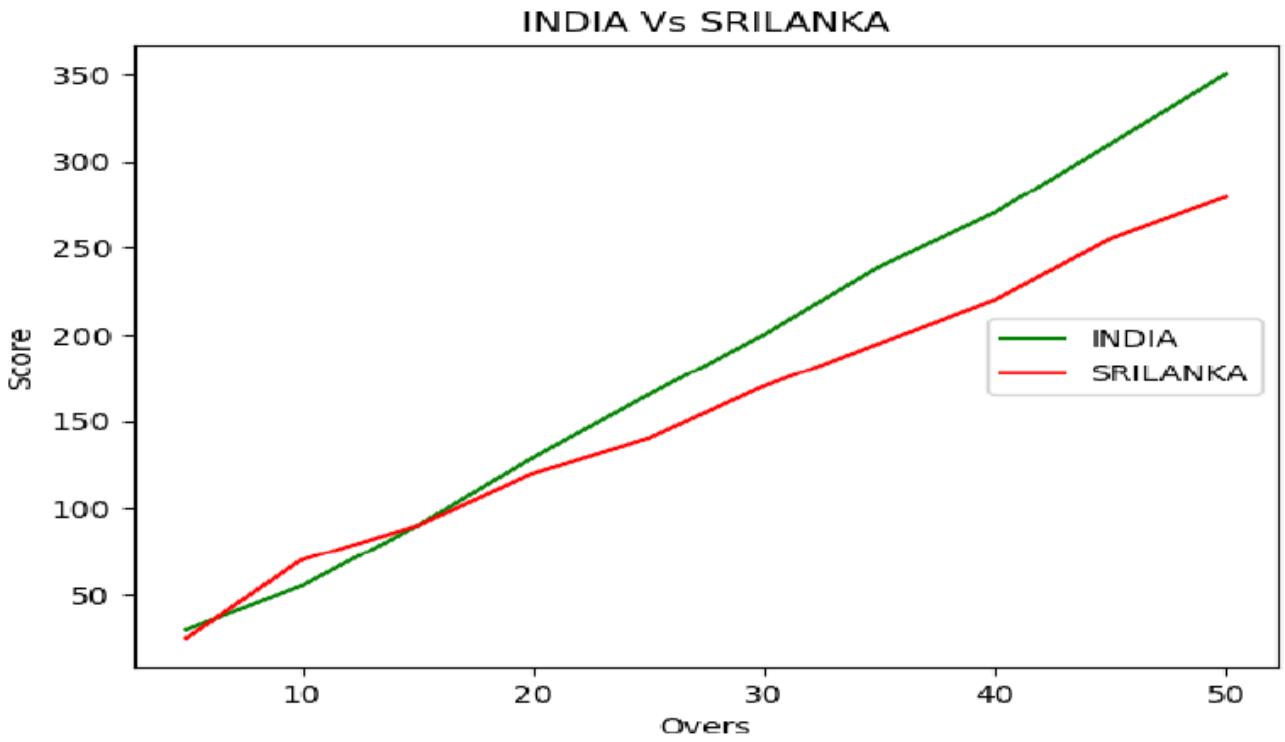
Aim

To visualize the given data using a line plot.

Source Code

```
import matplotlib.pyplot as cricket
Overs=list(range(5,51,5))
Indian_Score=[30,55,90,129,165,200,239,270,310,350]
Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]
cricket.plot(Overs,Indian_Score)
cricket.plot(Overs,Srilankan_Score)
cricket.show()
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend()
cricket.plot(Overs,Indian_Score,color="green",label="INDIA")
cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")
cricket.legend(loc="center right")
```

Output - Experiment 3.a



Result - Experiment 3.a

A line plot was created to visualize the supplied data.

Experiment 3.b: Show data visualization using a bar chart

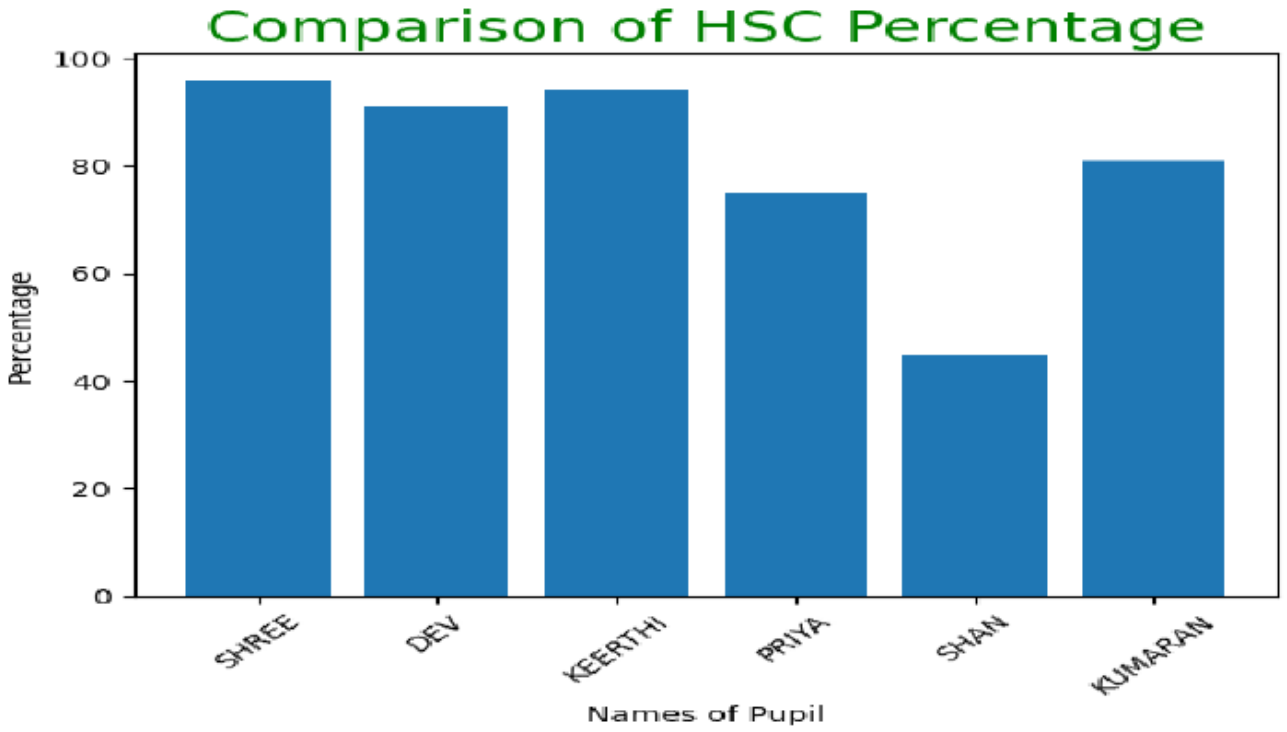
Aim

To visualize the given data using a bar chart.

Source Code

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

Output - Experiment 3.b



Result - Experiment 3.b

A bar chart was created to visualize the supplied data.

Experiment 3.c: Show data visualization using a pie chart

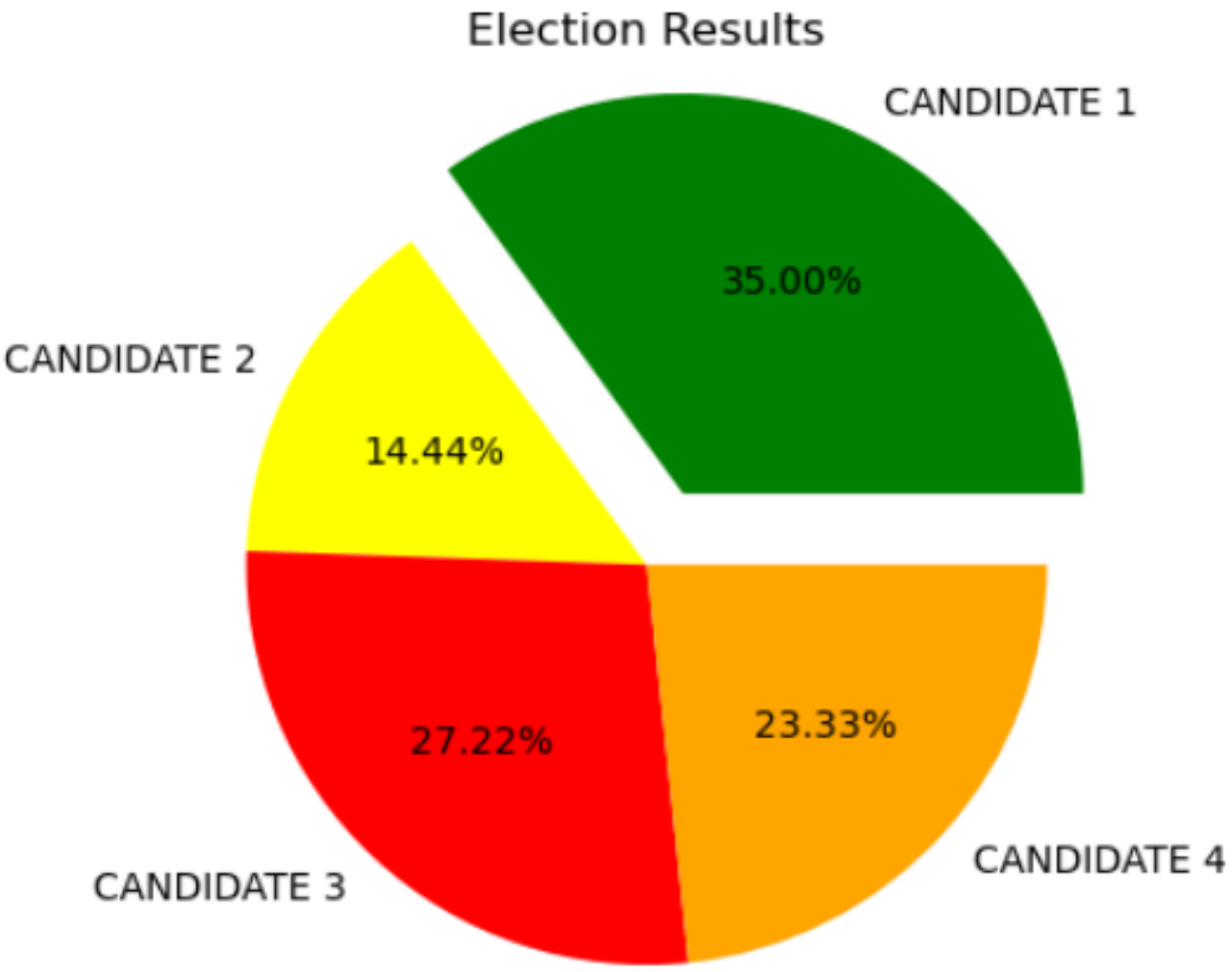
Aim

To visualize the given data using a pie chart.

Source Code

```
import matplotlib.pyplot as election
labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']
Votes = [315, 130, 245, 210]
colors = ['green', 'yellow', 'red', 'orange']
explode = (0.2, 0, 0, 0)
election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')
election.title('Election Results')
election.show()
```

Output - Experiment 3.c



Result - Experiment 3.c

A pie chart was created to visualize the supplied data.